

Element C

Presentation and justification of solution design requirements

Design requirements are listed and prioritized, and they are consistently clear and detailed; these design requirements presented are consistently objective, measurable, and they would be highly likely to lead to a tangible and viable solution to the problem identified; there is evidence that requirements represent the needs of, and have been validated by, many if not all primary stakeholder groups.

With the exception of a hook for a wall-mounted solution in the main enclosure, the Nashville Zoo Andean Bear keepers did not request for a specific solution or idea during the initial request. However, later in a follow up email, they responded to some questions of ours to help us best integrate our solution into the schedules of the bearkeepers.

Below are contents from the email with questions from our group italicized and answers from the zookeepers in red.

How often do the handlers enter the enclosure? Keeper staff will enter the habitat in the AM and have the ability to access in the afternoon if needed . As for their inside dens, they can access multiple times/day as needed.

How long do the bears need to be entertained (short and long-term)? The longer the duration the better. We prefer our bears to be engaged with the item as long as possible and its always great for them to return to it during different periods throughout the day.

Would you like the puzzle to be permanent or temporarily in the cage? We would like the enrichment device to be temporary. We would like to have the ability to move it to different places within the habitat as well as to their inside dens.

How do the handlers expect to administer the puzzle/extract the puzzle? The keepers will load the enrichment device with their diet and move it to the habitat or dens by themselves by hand.

What materials are safe for the bears/not safe? Bears are extremely strong animals so if you are making something from PVC, I would concentrate on schedule 40 or schedule 80 PVC. This material would be durable enough for them and easy to clean. Thick wood, 4x4, can also be used but it is not an easy to clean.

Do the puzzles need to be food-based (what incentivizes the bears other than food)? Bears are extremely food motivated so a majority of their enrichment devices revolve around food. They also like to push on items such as large car/truck tires.

Do the bears learn habits, if so how fast? Yes, they do learn habits and can develop one as quickly as a day.

Are we allowed to modify currently allocated resources (the current purple geometric toy inside the external cage for example)? We can provide you with this to modify if necessary.

What size should the largest portion of food be to fit inside the puzzle? We provided dry food items to the class when you were here last but the largest size that would be used would range from 2"x2" to 3".

In terms of dimensions, we decided that the maximum size of any possible design of the puzzle feeder is the size of a 31-gallon water drum. (22 inches in diameter, 35 inches tall). Is this a reasonable size? Yes! that's perfect!

How long are the Andean bears' attention span? It's all dependent on the bear and the day. If they are foraging for food via an enrichment device, they could be engaged for 5-10 minutes or until the food is depleted.

For reference, the engineering class currently has 3 groups working separately to build the puzzle feeders. In preference to the zookeepers, how many groups should be working on mounted feeder instead of a non-mounted feeder? All feeder devices that will be on habitat will need to be mounted one way or another. We will use either a chain or an eye bolt to secure. This is to ensure that the items do not make it into the moat, pool or hit the glass. Inside their dens, they are able to have an unmounted as well as a mounted device.

Can the feeder become detached from another portion of the feeder while inside the enclosure? We can not have the feeder become detached if it is on the exhibit.

From this feedback, we concluded to the following design requirements that fit that of the criteria of the needs of the Andean Bear Zookeepers.

The criterias for the Andean Bear Puzzle feeder should follow most or all of the following:

- A. Durability: Materials used should be sturdy and resilient to withstand the strength of the bears. PVC (schedule 40 or schedule 80) and thick wood (4x4) are recommended.
- B. Ease of Cleaning: Materials should be easy to clean to maintain hygiene standards.
- C. Temporary Design: The puzzle should be temporary to allow for movement within the habitat and dens.
- D. Feeding-based Enrichment: The puzzle should primarily revolve around food to incentivize the bears. However, other interactive elements like items to push can be included.
- E. Customization: The puzzle should be customizable to accommodate different sizes of food items, with the largest portion being 2"x2" to 3".
- F. Dimensions: The maximum size of the puzzle feeder should not exceed the dimensions of a 31-gallon water drum (22 inches in diameter, 35 inches tall).
- G. Mounting Requirement: Feeders for the habitat must be mounted securely to prevent them from falling into the moat, pool, or hitting the glass. Inside dens, options for both mounted and unmounted feeders are acceptable.
- H. Attention Span Consideration: The puzzle should engage the bears for as long as possible, ideally allowing them to return to it throughout the day. Attention spans vary but can range from 5-10 minutes or until food is depleted.

Criteria that should be highly prioritized are the following: A, B, C, D, G, and H. These criterias have been specifically or broadly mentioned by the zookeepers and serve a purpose as to benefit either the zookeepers, or bears, or both.

A. Durability

Using durable materials such as schedule 40 or schedule 80 PVC and thick wood (4x4) ensures the enrichment device withstands the Andean bears' strength, preventing breakage and potential injury. For zookeepers, this durability means fewer repairs or replacements, reducing maintenance workload and costs. Ensuring the device's longevity and safety is crucial, given the bears' strong physical capabilities.

B. Ease of Cleaning

Materials that are easy to clean help maintain hygiene, preventing bacteria buildup and ensuring a healthy environment for the bears. For zookeepers, easy-to-clean materials reduce the time and effort required for cleaning, allowing more efficient time management. Maintaining health standards is vital in a zoo setting to prevent exposure to harmful bacteria.

C. Temporary Design

A temporary design allows the puzzle to be moved within the habitat and dens, providing mental stimulation through environmental changes for the bears. For zookeepers, this flexibility ensures the bears can be engaged in different areas, making enrichment more effective and manageable. Preventing habituation by changing the device's location is key to maintaining the bears' interest.

D. Feeding-based Enrichment

Food-based enrichment keeps the bears engaged and stimulated, promoting natural foraging behaviors. For zookeepers, such devices ensure the bears are entertained for longer periods, reducing stress and promoting natural behaviors. Utilizing food as motivation aligns with the bears' natural tendencies and supports their cognitive development.

E. Customization

Customizable puzzles accommodate various food sizes, adding variety to the bears' diet and enrichment activities. For zookeepers, this flexibility allows easy adaptation to the bears' dietary needs and preferences, enhancing the effectiveness of enrichment. Providing a range of food sizes keeps the enrichment device interesting and challenging.

F. Dimensions

Ensuring the puzzle feeder is within a manageable size prevents overcrowding in the enclosure and allows comfortable interaction for the bears. For zookeepers, specified dimensions make the device easier to handle, move, and install, fitting well within the enclosure spaces. Adhering to practical dimensions ensures safety and ease of use.

G. Mounting Requirement

Securely mounted feeders prevent accidents, such as the device falling into water features or breaking glass, ensuring a safe environment for the bears. For zookeepers, this reduces the risk of accidents and potential injuries, creating a safer working environment and minimizing emergency interventions. Proper mounting is essential to prevent harm within the habitat.

H. Attention Span Consideration

Designing puzzles that engage bears for extended periods supports their cognitive and behavioral needs, preventing boredom. For zookeepers, longer engagement times with enrichment devices mean less frequent need for additional enrichment, allowing more efficient time management. Considering attention span ensures the device maintains the bears' interest and provides ongoing mental stimulation.